Event-Nodes and Texture-Nodes: Extending Causal Mechanistic Interpretability with Event-Historical Structure

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Abstract

Recent work in mechanistic interpretability has advanced a rigorously causal framework for understanding neural networks by treating internal components as variables within an interventional causal model. While this paradigm succeeds in distinguishing correlation from mechanism and enables algorithmic abstraction via counterfactual testing, it remains implicitly state-centric and only weakly sensitive to irreversibility and history.

This essay introduces a minimal extension to the causal–mechanistic framework by distinguishing two types of causal nodes: *texture-nodes*, which carry instantaneous values and are exhaustively characterized by value substitution, and *event-nodes*, which record irreversible structural commitments that constrain the space of future admissible states. Using a procedural generation analogy, texture-nodes correspond to parameter values of a generator, while event-nodes correspond to the selection of the generator itself.

We show that this distinction preserves the full rigor of causal abstraction while explaining substitution failure, semantic incompatibility, algorithmic mixtures, and structural irreversibility without invoking metaphysical assumptions. Mechanistic understanding is reframed as the reconstruction of both value-level mediation and event-level constraint structure under intervention.

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1 Introduction

Mechanistic interpretability seeks to answer a precise question: what internal structures of a neural network are responsible for its behavior, and in what sense? A purely observational approachexamining activations, attention maps, or weight magnitudesfails to answer this question, as correlation alone cannot establish causal responsibility.

Recent work has therefore adopted a causal paradigm, treating neural networks as systems amenable to intervention, counterfactual reasoning, and algorithmic abstraction. Under this view, understanding a model mechanistically requires identifying internal components whose manipulation predictably alters behavior in ways consistent with a hypothesized computation.

However, this paradigm remains implicitly committed to a state-based ontology: interventions act on instantaneous variable values, and history matters only insofar as it affects present behavior. This essay argues that such a framework is incomplete. Certain causal phenomenairreversibility, commitment, substitution failure, and algorithmic regime shiftscannot be fully explained without treating events themselves as first-class causal objects.

We propose a minimal extension: a typed causal graph distinguishing *texture-nodes* from *event-nodes*.

2 Causal Mechanistic Interpretability

In the narrow technical sense, mechanistic interpretability is defined by causal explanatory adequacy. A component is mechanistically relevant if intervening on it changes model behavior in a systematic and predictable way.

Interventions are typically implemented as value substitutions: fixing an internal activation to a value it would not naturally assume, or replacing it with the value observed under a different input. Counterfactual success is measured by alignment between low-level interventions and high-level algorithmic predictions.

This methodology supports *causal abstraction*: if interventions on low-level variables produce the same counterfactual outcomes as interventions on a high-level algorithm, the algorithm is considered an abstraction of the network.

Despite its power, this framework leaves open questions concerning history, irreversibility, and the conditions under which substitution itself is well defined.

3 Texture-Nodes: Value-Carrying Variables

Definition 1 (Texture-Node)**.** *A* texture-node *is a causal variable whose role is exhaustively determined by the value it instantiates at a given time.*

Formally, a texture-node corresponds to a variable $x_t \in \Sigma$, where causal influence is evaluated by substituting $x_t \mapsto x'_t$ and observing downstream effects.

Texture-nodes include neurons, attention heads, MLP activations, and linear directions used in activation steering. Their causal relevance is established through value-level interventions and mediation analysis.

Texture-nodes admit reversible manipulation: if the original value can be restored, the system returns to its prior functional state.

4 Event-Nodes: Constraint-Carrying Commitments

Definition 2 (Event-Node)**.** *An* event-node *is a causal variable whose role is not reducible to an instantaneous value, but consists in imposing a structural constraint on the space of admissible future states.*

An event-node induces a transformation

$$\mathcal{F}_{t+1} \subseteq \mathcal{F}_t,$$

where $\mathcal{F}_t$ denotes the set of reachable futures from time t.

Unlike texture-nodes, event-nodes are not exhaustively characterized by value substitution. Their causal influence is evaluated by changes in what the system can or cannot subsequently do.

Event-nodes are generally irreversible: no sequence of later texture-node interventions can restore the original future space once a structural commitment has been made.

5 Generator Functions and Procedural Structure

The distinction between event-nodes and texture-nodes can be clarified through a procedural generation analogy.

Let $\mathcal{G}$ be a family of generator functions, each parameterized by a vector $\theta \in \mathbb{R}^n$. A generator $G \in \mathcal{G}$ defines a family of outputs

$$y = G(\theta).$$

Texture-nodes correspond to the parameters θ or to ratios θ_i/θ_j, which function as vector-like directions in parameter space. Adjusting these values modifies surface appearance without changing the generative regime.

Event-nodes correspond to the selection of the generator G itself. Once a generator is chosen, all admissible parameterizations are conditioned on that choice. No amount of parameter tuning reproduces the output family of a different generator.

This distinction explains why some interventions modulate behavior smoothly, while others produce qualitative regime shifts.

6 Irreversibility and History

Standard causal analysis treats irreversibility as a consequence of information loss. Event-nodes introduce a second, more fundamental notion.

Definition 3 (Structural Irreversibility)**.** *An intervention is* structurally irreversible *if it reduces the space of admissible futures in a way that cannot be undone by subsequent value-level interventions.*

Structural irreversibility does not depend on signal destruction. It arises from commitment to a generative structure rather than from noise or ablation.

This explains why two systems with identical current states may behave differently under future interventions: they may share texture-state while differing in event-history.

7 Causal Abstraction with Typed Nodes

A causal abstraction must now satisfy two criteria:

1. **Texture Correspondence**: value-level interventions align between the abstraction and the underlying system.

2. **Event Correspondence**: structural commitments and irreversibility relations are preserved.

Multiple abstractions may remain valid at different levels, but abstractions that erase event-level structure mischaracterize the systems causal organization, even if short-horizon input–output behavior is preserved.

Algorithmic mixtures and regime switching are naturally interpreted as event-node transitions rather than as noise or failure.

8 Failure, Substitution, and Semantic Incompatibility

Interchange interventions assume semantic compatibility between substituted values and their new contexts. Event-nodes explain why this assumption sometimes fails.

Substitution failure does not indicate a mysterious semantic breakdown; it indicates that the substituted value was generated under a different event history and is incompatible with the current future constraints.

Novel behavior under intervention signals an incorrect or incomplete causal graph, often due to unmodeled event-nodes.

9 What Counts as Understanding

Mechanistic understanding is not merely the ability to control or predict behavior. It consists in reconstructing:

- how values mediate effects (texture-level),
- which commitments constrain futures (event-level),
- and how these interact under intervention.

Understanding is necessarily partial and revisable. No single abstraction exhausts the systems causal structure, but some abstractions fail by erasing irreversibility and history.

10 Conclusion

By distinguishing texture-nodes from event-nodes, we obtain a minimal extension to causal mechanistic interpretability that preserves its empirical rigor while accounting for irreversibility, history sensitivity, and structural commitment.

This extension requires no new metaphysics and no abandonment of causal abstraction. It merely clarifies what interventions act upon and what kinds of causal structure must be preserved for an explanation to remain valid.

Mechanistic interpretability, properly extended, becomes not only the study of what internal variables do, but of what systems have irreversibly become.